Nenad Petrovic, Vahid Zolfaghari, Andre Schamschurko, Sven Kirchner, Fengjunjie Pan, Chengdng Wu, Nils Purschke, Aleksei Velsh, Krzysztof Lebioda, Yinglei Song, Yi Zhang, Lukasz Mazur, and Alois Knoll. Survey of genai for automotive software development: From requirements to executable code, 2025. URL https://arxiv.org/abs/2507.15025.

Grzegorz Piotrowski, Mateusz Bystroński, Mikołaj Hołysz, Jakub Binkowski, Grzegorz Chodak, and Tomasz Jan Kajdanowicz. When will the tokens end? graph-based forecasting for LLMs output length. In Jin Zhao, Mingyang Wang, and Zhu Liu (eds.), *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pp. 843–848, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-254-1. doi: 10.18653/v1/2025.acl-srw.61. URL https://aclanthology.org/2025.acl-srw.61/.

Haoran Qiu, Weichao Mao, Archit Patke, Shengkun Cui, Saurabh Jha, Chen Wang, Hubertus Franke, Zbigniew T. Kalbarczyk, Tamer Başar, and Ravishankar K. Iyer. Efficient interactive llm serving with proxy model-based sequence length prediction. In *The 5th International Workshop on Cloud Intelligence / AIOps at ASPLOS 2024*, volume 5, pp. 1–7, San Diego, CA, USA, 2024. Association for Computing Machinery.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. In *Advances in neural information processing systems*, 2019.

Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. Agent laboratory: Using llm agents as research assistants, 2025. URL https://arxiv.org/abs/2501.04227.

Uri Shaham, Maor Ivgi, Avia Efrat, Jonathan Berant, and Omer Levy. Zeroscrolls: A zero-shot benchmark for long text understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 7977–7989, 2023.

Rana Shahout, Eran Malach, Chunwei Liu, Weifan Jiang, Minlan Yu, and Michael Mitzenmacher. Don't stop me now: Embedding based scheduling for llms. In *The Thirteenth International Conference on Learning Representations*, 2025.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL https://arxiv.org/abs/2402.03300.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks, 2017. URL https://arxiv.org/abs/1703.01365.

Liangyu Wang, Huanyi Xie, Xinhai Wang, Tianjin Huang, Mengdi Li, and Di Wang. Infinite sampling: Efficient and stable grouped rl training for large language models. *arXiv preprint arXiv:2506.22950*, 2025a.

Liangyu Wang, Huanyi Xie, Xinhai Wang, Tianjin Huang, Mengdi Li, and Di Wang. Infinite sampling: Efficient and stable grouped rl training for large language models, 2025b. URL https://arxiv.org/abs/2506.22950.

Mo Xuan, Zhang yue, and Wu Weigang. Maaso: Slo-aware orchestration of heterogeneous model instances for maas, 2025. URL https://arxiv.org/abs/2509.06362.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng,

Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025a. URL https://arxiv.org/abs/2505.09388.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025b. URL https://arxiv.org/abs/2412.15115.

Junchi Yao, Jianhua Xu, Tianyu Xin, Ziyi Wang, Shenzhe Zhu, Shu Yang, and Di Wang. Is your llm-based multi-agent a reliable real-world planner? exploring fraud detection in travel planning. *arXiv preprint arXiv:2505.16557*, 2025.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, Wang Zhang, Hang Zhu, Jinhua Zhu, Jiaze Chen, Jiangjie Chen, Chengyi Wang, Hongli Yu, Yuxuan Song, Xiangpeng Wei, Hao Zhou, Jingjing Liu, Wei-Ying Ma, Ya-Qin Zhang, Lin Yan, Mu Qiao, Yonghui Wu, and Mingxuan Wang. Dapo: An open-source llm reinforcement learning system at scale, 2025. URL https://arxiv.org/abs/2503.14476.

Sungmin Yun, Kwanhee Kyung, Juhwan Cho, Jaewan Choi, Jongmin Kim, Byeongho Kim, Sukhan Lee, Kyomin Sohn, and Jung Ho Ahn. Duplex: A device for large language models with mixture of experts, grouped query attention, and continuous batching, 2024. URL https://arxiv.org/abs/2409.01141.

Jiaming Zhang, Mingxi Lei, Meng Ding, Mengdi Li, Zihang Xiang, Difei Xu, Jinhui Xu, and Di Wang. Towards user-level private reinforcement learning with human feedback. *arXiv preprint arXiv:2502.17515*, 2025.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

Zhuoran Zhang, Yongxiang Li, Zijian Kan, Keyuan Cheng, Lijie Hu, and Di Wang. Locate-then-edit for multi-hop factual recall under knowledge editing. *arXiv preprint arXiv:2410.06331*, 2024.

Ranran Zhen, Juntao Li, Yixin Ji, Zhenlin Yang, Tong Liu, Qingrong Xia, Xinyu Duan, Zhefeng Wang, Baoxing Huai, and Min Zhang. Taming the titans: A survey of efficient llm inference serving, 2025. URL https://arxiv.org/abs/2504.19720.

Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, Jingren Zhou, and Junyang Lin. Group sequence policy optimization, 2025. URL https://arxiv.org/abs/2507.18071.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric Xing, et al. Lmsys-chat-1m: A large-scale real-world llm conversation dataset. In *The Twelfth International Conference on Learning Representations*, 2024.

Zangwei Zheng, Xiaozhe Ren, Fuzhao Xue, Yang Luo, Xin Jiang, and Yang You. Response length perception and sequence scheduling: An llm-empowered llm inference pipeline. In *Advances in Neural Information Processing Systems*, volume 36, pp. 65517–65530, 2023.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

Appendix

A THE USE OF LARGE LANGUAGE MODELS(LLMS)

During the preparation of this manuscript, we utilized Large Language Models (LLMs) for assistance with grammar checking and text polishing to enhance the clarity and readability of the paper.

B EVALUATION METRICS

This section details the metrics used to evaluate the performance of our models and the end-to-end system.

B.1 PREDICTOR PERFORMANCE METRIC

To evaluate the performance of our predictor, we use the **Mean Absolute Error (MAE)**.

- **Mean Absolute Error (MAE)** measures the average absolute difference between the predicted values ($\hat{y}_i$) and the actual ground truth values (y_i). A lower MAE indicates a more accurate prediction model. Given n samples, it is defined as:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

B.2 END-TO-END SYSTEM PERFORMANCE METRICS

To evaluate the overall system performance, we use throughput, Job Completion Time, and padding ratio.

- **Throughput** measures the number of jobs the system can successfully process per unit of time. It is a key indicator of system efficiency. Let $N_{\text{completed}}$ be the total number of completed jobs in a time interval T_{total}:

$$\text{Throughput} = \frac{N_{\text{completed}}}{T_{\text{total}}}$$

- **Job Completion Time (JCT)** refers to the total time elapsed from a job's submission ($T_{\text{submission}}$) to its completion ($T_{\text{completion}}$). We typically measure the average JCT across all jobs to gauge system responsiveness.

$$\text{JCT} = T_{\text{completion}} - T_{\text{submission}}$$

- **Padding Ratio** quantifies the overhead or resource waste from padding. It is the ratio of the padded portion of a resource to the actual required resource size (R_{actual}), where $R_{\text{allocated}}$ is the total allocated resource. A lower ratio is better.

$$\text{Padding Ratio} = \frac{R_{\text{allocated}} - R_{\text{actual}}}{R_{\text{actual}}}$$

C BENCHMARK DETAILS

C.1 DATASET STATISTICS